

Program : Diploma in Polymer Technology	
Course Code : 3098	Course Title: Physical Testing Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0,P:3)	Periods per semester: 45

Course Objectives:

- To develop skill for determination of physical properties of various rubber compounds

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge in Mathematics		Mathematics I	1
Basic knowledge in Chemistry		Applied Chemistry	1
Basic knowledge in Physics		Applied Physics	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Prepare the rubber compound and determine the of specific gravity and curing properties of rubber compounds	12	Applying
CO2	Prepare test specimens for determination of mechanical properties	8	Applying
CO3	Determine the mechanical properties of vulcanized rubber samples	23	Applying
	Lab exam	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3	3		3			
CO3	3	3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Prepare rubber compounds and determine the specific gravity and cure characteristics		
M1.01	Prepare rubber compounds for test specimen	6	Applying
M1.02	Determine the specific gravity of rubber compounds	3	Applying
M1.03	Determine the cure characteristics of rubber compounds	3	Applying
CO2	Prepare the test specimen for determination of mechanical properties		
M2.01	Prepare the test specimen by moulding process.	5	Applying
M2.02	Prepare the test specimen and determine mechanical properties.	3	Applying
	Lab exam I	1	
CO3	Determine the mechanical properties of vulcanised rubber samples		
M3.01	Determine the hardness of rubber vulcanisate.	3	Applying
M3.02	Determine the tensile properties of rubber vulcanisate.	3	Applying
M3.03	Determine the tear strength of rubber vulcanisate.	3	Applying
M3.04	Determine the accelerated ageing properties of rubber vulcanisate.	3	Applying
M3.05	Determine the tension set of rubber vulcanisate.	3	Applying
M3.06	Determine the compression set of rubber vulcanisate.	3	Applying
M3.07	Determine the resilience of rubber vulcanisate.	3	Applying

M3.08	Determine the abrasion resistance of rubber vulcanisate.	2	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
T1	Hand book of polymer testing-RP BROWN
R2	ASTM Volumes I,II,III
R3	Physical testing of rubbers- RP brown

Online Resources:

SL.No	Website Link
1	https://www.researchgate.net
2	https://www.tainstruments.com